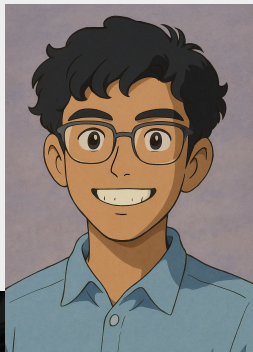


Beyond Traditional Forecasting:

Merging Gold Market Data with News Sentiment

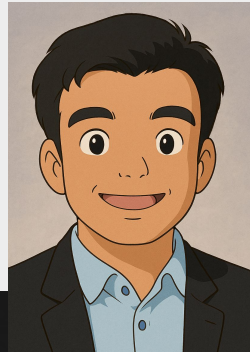
MEET THE TEAM



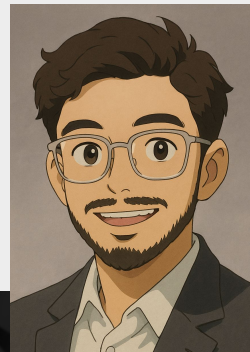
Arko
Bhattacharya



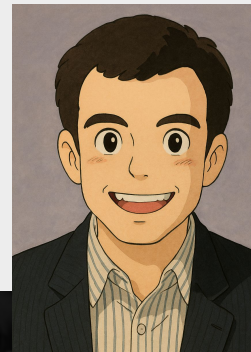
Zachary
Fennie



Vishesh
Gupta



Danish
Maknojia



Chris
Moreira

Imagine...

A new administration implements reciprocal tariffs targeting countries with trade deficits

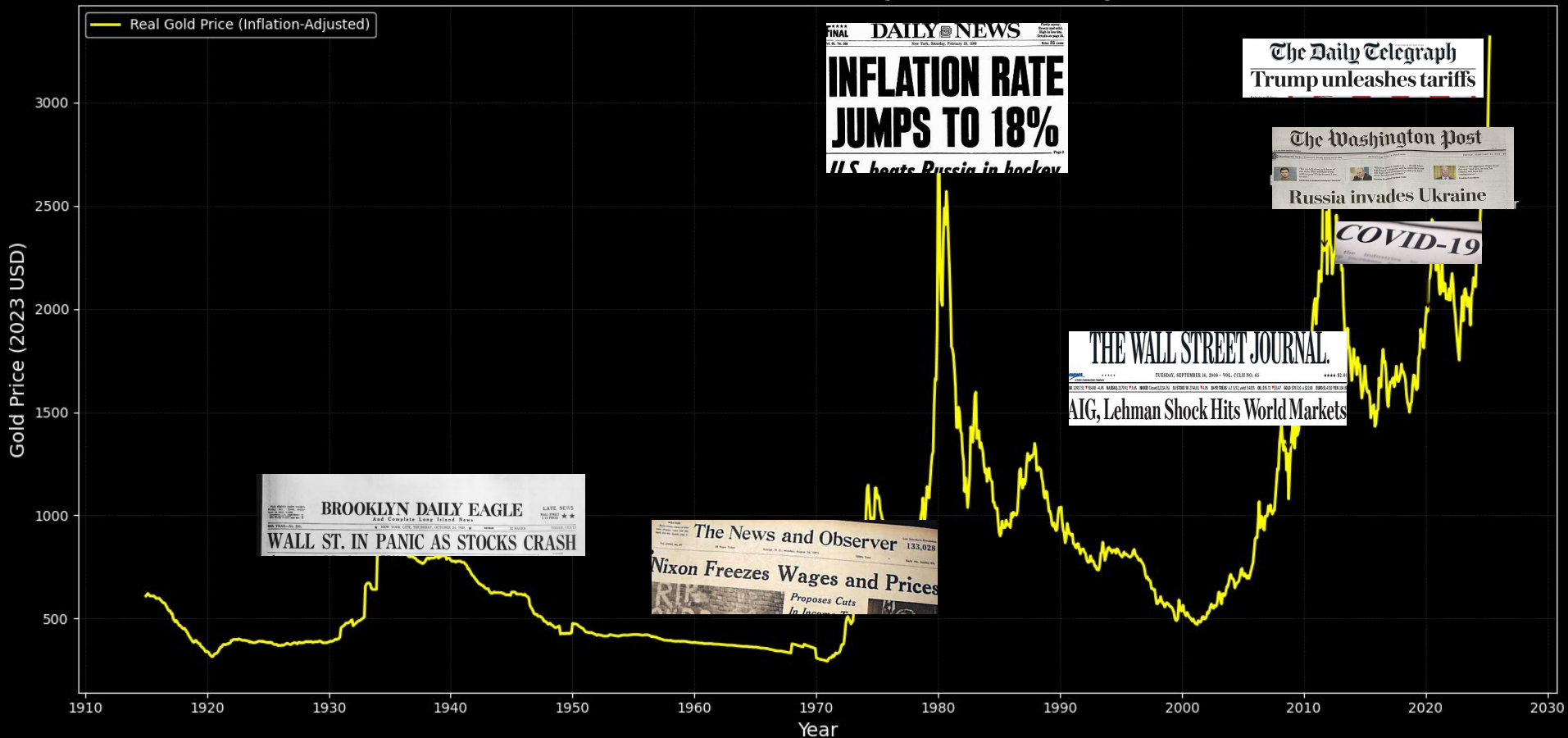
Tariffs vary significantly, ranging from 20% to over 80%, causing higher prices, market volatility, and recession risks

So, in all the chaos, where should you invest your money...



GOLD

Real Gold Price in USD (1915-Present)



AlphaGen Framework

BASELINE

Base LSTM model
architecture
trained on opening
prices

1

SENTIMENT

Supervised
sentiment scoring
on related news
headlines
(FinBERT)

2

CLUSTER

Unsupervised
clustering on news
headlines (BERTopic)
+
LLM Inference on
cluster topics (Gemini
2.0 Flash)

3

COMBINED

Added aggregated
NLP features to
baseline model
+
Real-Time
Prediction
Pipeline

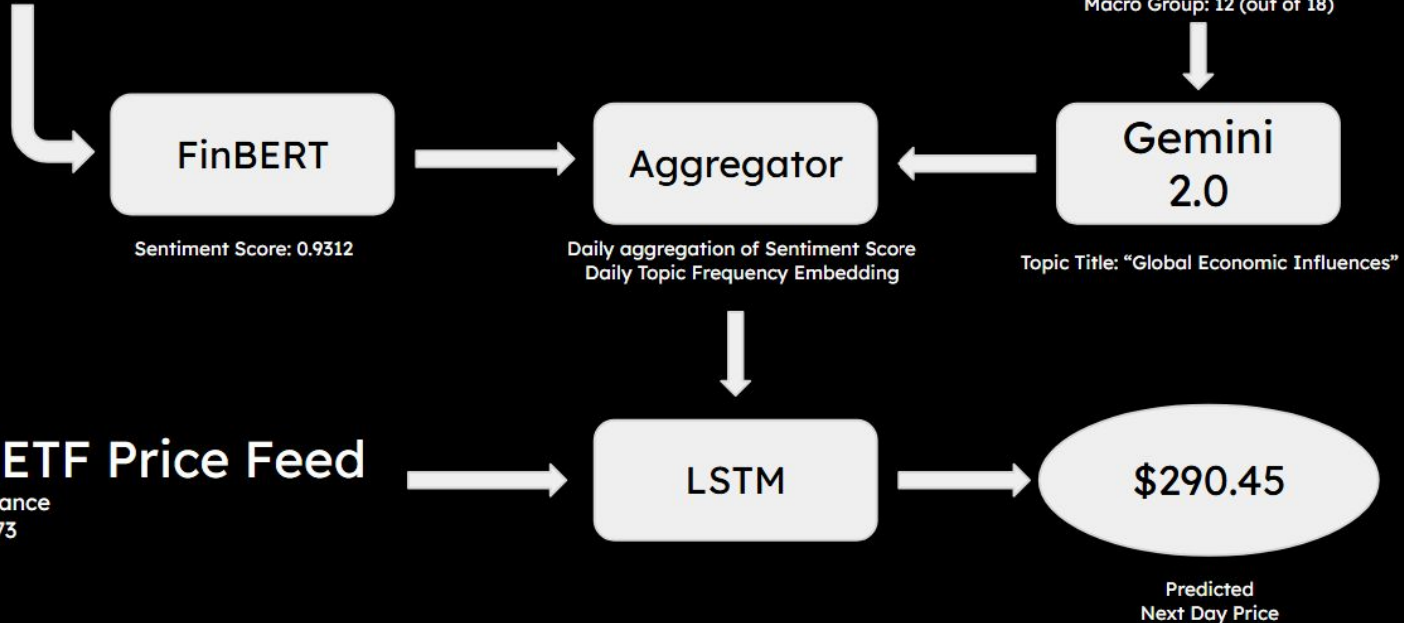
4

Realtime Flow

News Headline Stream

Source: newsdata.io

Sample Headline: "Gold surges as US inflation hits 5%"



Why Directional Accuracy?

“In markets, being right just slightly more than wrong can make all the difference.” - Warren Buffett

Issue with Traditional Metrics

- Our model, when applied to price data, often exhibits over-smoothing
- This leads to muted predictions that fail to capture real market dynamics

Directional Accuracy: A Better Metric

- Measures how often the model correctly predicts the direction (up or down).
- More useful for decision-making (e.g., investment/trading signals)
- Robust to over-smoothing, as it focuses on movement correctness



Results

Performance improves progressively with the addition of new feature.

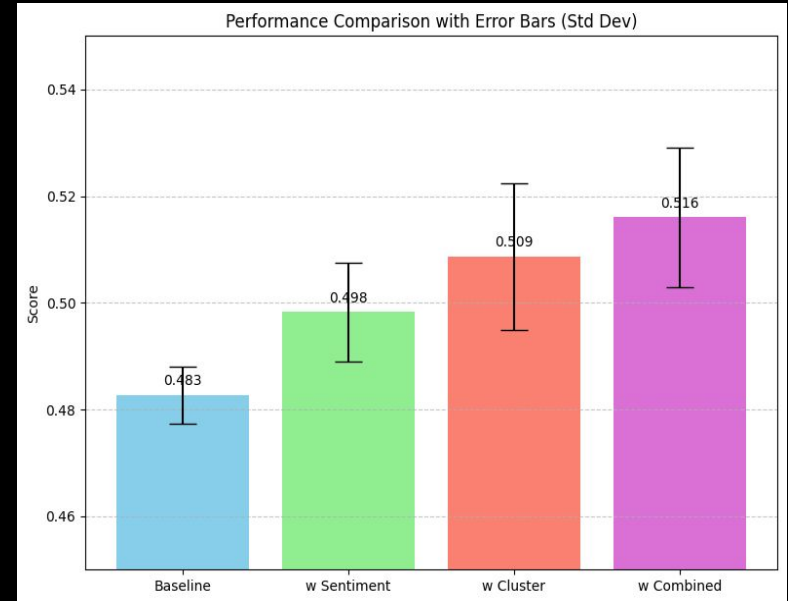
Standard deviation bars indicate the variability with added features however the trend shows consistent performance gains.

So, what were our real time results ?

Model predicted 28th April 2025 gold ETF prices at: \$313.9

Actual Price: \$309.7, previous day: \$304.2

Direction Trend Prediction: Upward which is correct



What's Next?

01

Existing Framework

Current predictive pipeline with thematic and sentiment approach

Multilingual Sentiment Layer

Integrate Global News Sources & Multilingual Models (e.g., mBERT, XLM-R)

02

03

Technical Indicators

Integrate Technical Indicators (e.g., RSI, Bollinger Bands, Moving Averages) to enrich signal generation

Real-time Trading Algorithm

Deploy Real-Time Trading Strategy that will execute trades in the gold market

04



Q&A